THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 22	Arc $= \frac{\theta}{360} \times 2\pi r$.
2	**(c)** 12	$h = \frac{2A}{b}$.
3	**(d)** 12	$h = \frac{A}{b}$.
4	**(d)** 52	Side $= \sqrt{A}$.
5	**(d)** 140	Area $= \frac{1}{2}d(h_1 + h_2)$.
6	**(a)** 126	Find the breadth first.
7	**(a)** 11	Perimeter is unchanged.
8	**(c)** 110	Distance $= n\times$ circumference.
9	**(c)** 88	Arc $= \frac{\theta}{360} \times 2\pi r$.
10	**(d)** 8	$h = \frac{2A}{b}$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$s = 32$, $A = 168$ cm^2 (1); $h = \frac{2\times168}{14} = 24$ cm (1).	1+1
12	$A = \frac{1}{2} \times 26 \times 9 + \frac{1}{2} \times 26 \times 7$ (1); $A = 208$ cm^2 (1).	1+1
13	Arc $= \frac{40}{360} \times 2 \times \frac{22}{7} \times 63$ (1); $= 44$ cm (1).	1+1
14	$s = 27$, $A = 126$ cm^2 (1); $h = \frac{2\times126}{21} = 12$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$s = 20$ cm (1); $s-a = 16$, $s-b = 12$, $s-c = 9$, $s-d = 3$ (1); $A = 72$ cm^2 (1).	1+1+1
16	Perimeter $= 2 \times 30 + 14 + \frac{1}{2}\pi \times 14 = 74 + 22$; $= 96$ cm (1); Area $= 420 + \frac{1}{2}\pi \left(\frac{14}{2}\right)^2$ (1); $= 497$ cm^2 (1).	1+1+1
17	$A = 15 \times 8 = 120$ cm^2 (1); $h' = \frac{120}{12} = 10$ cm (2).	1+2
	OR: $s = 24$ cm (1); $s-a = 16$, $s-b = 15$, $s-c = 12$, $s-d = 5$ (1); $A = 120$ cm^2 (1).	1+1+1
18	Perimeter $= 2 \times 25 + 28 + \frac{1}{2}\pi \times 28 = 78 + 44$; $= 122$ cm (1); Area $= 700 + \frac{1}{2}\pi \left(\frac{28}{2}\right)^2$ (1); $= 1008$ cm^2 (1).	1+1+1
	OR: $A = 18 \times 5 = 90$ cm^2 (1); $h' = \frac{90}{9} = 10$ cm (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Plot area $= 1200$ m^2 (1); Pond area $= \frac{22}{7} \times 7^2 = 154$ m^2 (1); Remaining $= 1046$ m^2 (1); Cost $= 1046 \times 6$ (1); $=$ ₹6276 (1).	1+1+1+1+1
	OR: $s = 18$ m (1); $A = \sqrt{s(s-a)(s-b)(s-c)} = 60$ m^2 (1); Cost $= 60 \times 5$ (1); $=$ ₹300 (1); Units stated (1).	1+1+1+1+1
20	Plot area $= 2000$ m^2 (1); Pond area $= \frac{22}{7} \times 14^2 = 616$ m^2 (1); Remaining $= 1384$ m^2 (1); Cost $= 1384 \times 4$ (1); $=$ ₹5536 (1).	1+1+1+1+1
	OR: $s = 27$ m (1); $A = \sqrt{s(s-a)(s-b)(s-c)} = 108$ m^2 (1); Cost $= 108 \times 4$ (1); $=$ ₹432 (1); Units stated (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up circumference ($2\pi r$) and area (πr^2), or writing the area in cm instead of cm^2.
- In Heron's formula, using the perimeter instead of the semi-perimeter s in the factors $(s-a)$.
- Using a slanting side instead of the perpendicular height in the area of a parallelogram or triangle.
- Treating the width of a path as the outer radius; the outer radius is $r + w$ and the area of the path is the *difference* of the two circle areas.
- Forgetting the factor $\frac{\theta}{360}$ in the arc length formula, or using the area formula πr^2 for arc length.
- Applying Brahmagupta's formula to a quadrilateral that is not cyclic.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 8	$h = \frac{A}{b}$.
2	**(a)** 72	Side $= \sqrt{A}$.
3	**(d)** 108	Area $= \frac{1}{2}d(h_1 + h_2)$.
4	**(b)** 198	Find the breadth first.
5	**(a)** 33	Perimeter is unchanged.
6	**(a)** 44	Distance $= n\times$ circumference.
7	**(c)** 22	Arc $= \frac{\theta}{360} \times 2\pi r$.
8	**(a)** 24	$h = \frac{2A}{b}$.
9	**(d)** 14	$h = \frac{A}{b}$.
10	**(d)** 96	Side $= \sqrt{A}$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$A = \frac{1}{2} \times 30 \times 8 + \frac{1}{2} \times 30 \times 12$ (1); $A = 300$ cm^2 (1).	1+1
12	Arc $= \frac{72}{360} \times 2 \times \frac{22}{7} \times 35$ (1); $= 44$ cm (1).	1+1
13	$s = 27$, $A = 108$ cm^2 (1); $h = \frac{2\times108}{24} = 9$ cm (1).	1+1
14	$A = \frac{1}{2} \times 22 \times 10 + \frac{1}{2} \times 22 \times 6$ (1); $A = 176$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$s = 28$ cm (1); $s - a = 24$, $s - b = 18$, $s - c = 8$, $s - d = 6$ (1); $A = 144$ cm^2 (1).	1+1+1
16	Perimeter $= 2 \times 40 + 14 + \frac{1}{2}\pi \times 14 = 94 + 22$; $= 116$ cm (1); Area $= 560 + \frac{1}{2}\pi \left(\frac{14}{2}\right)^2$ (1); $= 637$ cm^2 (1).	1+1+1
17	$A = 20 \times 12 = 240$ cm^2 (1); $h' = \frac{240}{16} = 15$ cm (2).	1+2
	OR: $s = 30$ cm (1); $s - a = 25$, $s - b = 18$, $s - c = 9$, $s - d = 8$ (1); $A = 180$ cm^2 (1).	1+1+1
18	Perimeter $= 2 \times 35 + 28 + \frac{1}{2}\pi \times 28 = 98 + 44$; $= 142$ cm (1); Area $= 980 + \frac{1}{2}\pi \left(\frac{28}{2}\right)^2$ (1); $= 1288$ cm^2 (1).	1+1+1
	OR: $A = 24 \times 9 = 216$ cm^2 (1); $h' = \frac{216}{18} = 12$ cm (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Plot area $= 2100$ m^2 (1); Pond area $= \frac{22}{7} \times 14^2 = 616$ m^2 (1); Remaining $= 1484$ m^2 (1); Cost $= 1484 \times 5$ (1); $=$ ₹7420 (1).	1+1+1+1+1
	OR: $s = 32$ m (1); $A = \sqrt{s(s-a)(s-b)(s-c)} = 168$ m^2 (1); Cost $= 168 \times 3$ (1); $=$ ₹504 (1); Units stated (1).	1+1+1+1+1
20	Plot area $= 2025$ m^2 (1); Pond area $= \frac{22}{7} \times 7^2 = 154$ m^2 (1); Remaining $= 1871$ m^2 (1); Cost $= 1871 \times 8$ (1); $=$ ₹14968 (1).	1+1+1+1+1
	OR: $s = 34$ m (1); $A = \sqrt{s(s-a)(s-b)(s-c)} = 204$ m^2 (1); Cost $= 204 \times 2$ (1); $=$ ₹408 (1); Units stated (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up circumference ($2\pi r$) and area (πr^2), or writing the area in cm instead of cm^2.
- In Heron's formula, using the perimeter instead of the semi-perimeter s in the factors $(s - a)$.
- Using a slanting side instead of the perpendicular height in the area of a parallelogram or triangle.
- Treating the width of a path as the outer radius; the outer radius is $r + w$ and the area of the path is the *difference* of the two circle areas.
- Forgetting the factor $\frac{\theta}{360}$ in the arc length formula, or using the area formula πr^2 for arc length.
- Applying Brahmagupta's formula to a quadrilateral that is not cyclic.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 240	Area $= \frac{1}{2}d(h_1 + h_2)$.
2	**(b)** 260	Find the breadth first.
3	**(d)** 55	Perimeter is unchanged.
4	**(b)** 132	Distance $= n\times$ circumference.
5	**(d)** 88	Arc $= \frac{\theta}{360} \times 2\pi r$.
6	**(a)** 8	$h = \frac{2A}{b}$.
7	**(c)** 9	$h = \frac{A}{b}$.
8	**(b)** 128	Side $= \sqrt{A}$.
9	**(b)** 210	Area $= \frac{1}{2}d(h_1 + h_2)$.
10	**(c)** 360	Find the breadth first.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Arc $= \frac{240}{360} \times 2 \times \frac{22}{7} \times 21$ (1); $= 88$ cm (1).	1+1
12	$s = 33$, $A = 132$ cm^2 (1); $h = \frac{2\times132}{30} = \frac{44}{5}$ cm (1).	1+1
13	$A = \frac{1}{2} \times 28 \times 5 + \frac{1}{2} \times 28 \times 9$ (1); $A = 196$ cm^2 (1).	1+1
14	Arc $= \frac{150}{360} \times 2 \times \frac{22}{7} \times 42$ (1); $= 110$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$s = 32$ cm (1); $s-a = 28$, $s-b = 18$, $s-c = 14$, $s-d = 4$ (1); $A = 168$ cm^2 (1).	1+1+1
16	Perimeter $= 2 \times 21 + 14 + \frac{1}{2}\pi \times 14 = 56 + 22$; $= 78$ cm (1); Area $= 294 + \frac{1}{2}\pi \left(\frac{14}{2}\right)^2$ (1); $= 371$ cm^2 (1).	1+1+1
17	$A = 30 \times 10 = 300$ cm^2 (1); $h' = \frac{300}{25} = 12$ cm (2).	1+2
	OR: $s = 33$ cm (1); $s-a = 27$, $s-b = 25$, $s-c = 8$, $s-d = 6$ (1); $A = 180$ cm^2 (1).	1+1+1
18	Perimeter $= 2 \times 50 + 28 + \frac{1}{2}\pi \times 28 = 128 + 44$; $= 172$ cm (1); Area $= 1400 + \frac{1}{2}\pi \left(\frac{28}{2}\right)^2$ (1); $= 1708$ cm^2 (1).	1+1+1
	OR: $A = 14 \times 12 = 168$ cm^2 (1); $h' = \frac{168}{21} = 8$ cm (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Plot area $= 4000$ m^2 (1); Pond area $= \frac{22}{7} \times 21^2 = 1386$ m^2 (1); Remaining $= 2614$ m^2 (1); Cost $= 2614 \times 3$ (1); $=$ ₹7842 (1).	1+1+1+1+1
	OR: $s = 27$ m (1); $A = \sqrt{s(s-a)(s-b)(s-c)} = 90$ m^2 (1); Cost $= 90 \times 6$ (1); $=$ ₹540 (1); Units stated (1).	1+1+1+1+1
20	Plot area $= 2940$ m^2 (1); Pond area $= \frac{22}{7} \times 14^2 = 616$ m^2 (1); Remaining $= 2324$ m^2 (1); Cost $= 2324 \times 7$ (1); $=$ ₹16268 (1).	1+1+1+1+1
	OR: $s = 35$ m (1); $A = \sqrt{s(s-a)(s-b)(s-c)} = 210$ m^2 (1); Cost $= 210 \times 2$ (1); $=$ ₹420 (1); Units stated (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up circumference ($2\pi r$) and area (πr^2), or writing the area in cm instead of cm^2.
- In Heron's formula, using the perimeter instead of the semi-perimeter s in the factors $(s-a)$.
- Using a slanting side instead of the perpendicular height in the area of a parallelogram or triangle.
- Treating the width of a path as the outer radius; the outer radius is $r + w$ and the area of the path is the *difference* of the two circle areas.
- Forgetting the factor $\frac{\theta}{360}$ in the arc length formula, or using the area formula πr^2 for arc length.
- Applying Brahmagupta's formula to a quadrilateral that is not cyclic.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.